

Лабораторная работа №4

Работа с программными пакетами

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16 сентября 2025

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Цель работы

Получить навыки работы с репозиториями, менеджерами пакетов **dnf** и **rpm** в операционной системе Linux.

Ход выполнения работы

Работа с репозиториями DNF

```
root@rishardkogengar: /home/rkogengar#  
root@rishardkogengar: /home/rkogengar# cd /etc/yum.repos.d/  
root@rishardkogengar: /etc/yum.repos.d# ls  
rocky-addons.repo rocky-devel.repo rocky-extras.repo rocky.repo  
root@rishardkogengar: /etc/yum.repos.d# cat rocky.repo  
# rocky.repo  
#  
# The mirrorlist system uses the connecting IP address of the client and the  
# update status of each mirror to pick current mirrors that are geographically  
# close to the client. You should use this for Rocky updates unless you are  
# manually picking other mirrors.  
#  
# If the mirrorlist does not work for you, you can try the commented out  
# baseurl line instead.  
  
[baseos]  
name=Rocky Linux $releasever - BaseOS  
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=BaseOS-$releasever$rltype  
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/BaseOS/$basearch/os/  
gpgcheck=1  
enabled=1  
countme=1  
metadata_expire=6h  
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-Rocky-10  
  
[baseos-debuginfo]  
name=Rocky Linux $releasever - BaseOS - Debug  
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=BaseOS-$releasever-debug$rltype  
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/BaseOS/$basearch/debug/tree/  
gpgcheck=1  
enabled=0  
metadata_expire=6h
```

Работа с репозиториями DNF

```
root@rishardkogengar: /etc/yum.repos.d# dnf repolist
repo id                                repo name
appstream                              Rocky Linux 10 - AppStream
baseos                                 Rocky Linux 10 - BaseOS
extras                                 Rocky Linux 10 - Extras
root@rishardkogengar: /etc/yum.repos.d# dnf search user
Rocky Linux 10 - BaseOS                17 MB/s | 16 MB      00:00
Rocky Linux 10 - AppStream             4.4 MB/s | 2.1 MB   00:00
Rocky Linux 10 - Extras                6.9 kB/s | 4.9 kB   00:00
===== Name & Summary Matched: user =====
gnome-user-docs.noarch : GNOME User Documentation
libuser.x86_64 : A user and group account administration library
perl-User-pwent.noarch : By-name interface to Perl built-in user name resolver
samba-usershares.x86_64 : Provides support for non-root user shares
usermode.x86_64 : Tools for certain user account management tasks
userspace-rcu.x86_64 : RCU (read-copy-update) implementation in user-space
xdg-user-dirs.x86_64 : Handles user special directories
===== Name Matched: user =====
gnome-shell-extension-user-theme.noarch : Support for custom themes in GNOME Shell
xdg-user-dirs-gtk.x86_64 : Gnome integration of special directories
===== Summary Matched: user =====
NetworkManager.x86_64 : Network connection manager and user applications
PackageKit-command-not-found.x86_64 : Ask the user to install command line programs automatically
xorg-x11-fonts-75dpi.x86_64 : 75 DPI font set for medium and medium-high resolution screens
```

Рис. 2: Список репозиторийев

Поиск и установка пакетов

```
-----
root@rishardkogengar:/etc/yum.repos.d# dnf search nmap
Last metadata expiration check: 0:00:20 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
===== Name Exactly Matched: nmap =====
nmap.x86_64 : Network exploration tool and security scanner
===== Name & Summary Matched: nmap =====
nmap-ncat.x86_64 : Nmap's Netcat replacement
root@rishardkogengar:/etc/yum.repos.d# dnf info nmap
Last metadata expiration check: 0:00:29 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Available Packages
Name           : nmap
Epoch         : 4
Version        : 7.92
Release        : 3.el10
Architecture   : x86_64
Size           : 5.6 M
Source         : nmap-7.92-3.el10.src.rpm
Repository     : appstream
Summary        : Network exploration tool and security scanner
URL            : http://nmap.org/
License        : LicenseRef-Nmap
Description    : Nmap is a utility for network exploration or security auditing. It supports
                  : ping scanning (determine which hosts are up), many port scanning techniques
                  : (determine what services the hosts are offering), and TCP/IP fingerprinting
                  : (remote host operating system identification). Nmap also offers flexible target
                  : and port specification, decoy scanning, determination of TCP sequence
                  : predictability characteristics, reverse-identd scanning, and more. In addition
                  : to the classic command-line nmap executable, the Nmap suite includes a flexible
                  : data transfer, redirection, and debugging tool (netcat utility ncat), a utility
                  : for comparing scan results (ndiff), and a packet generation and response
                  : analysis tool (nping).

root@rishardkogengar:/etc/yum.repos.d# █
```

```
-----
Total                               12 MB/s | 5.6 MB    00:00
Rocky Linux 10 - AppStream          1.5 MB/s | 1.6 kB    00:00
Importing GPG key 0x6FEDFC85:
  Userid      : "Release Engineering (Rocky Linux 10) <releng@rockylinux.org>"
  Fingerprint: FC22 6859 C086 0BF0 DDB9 5B08 5B10 6C73 6FED FC85
  From        : /etc/pki/rpm-gpg/RPM-GPG-KEY-Rocky-10
Is this ok [y/N]: y
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : nmap-4:7.92-3.el10.x86_64      1/1
  Running scriptlet: nmap-4:7.92-3.el10.x86_64      1/1

Installed:
  nmap-4:7.92-3.el10.x86_64

Complete!
root@rishardkogengar:/etc/yum.repos.d# dnf install nmap\*
Last metadata expiration check: 0:01:18 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Package nmap-4:7.92-3.el10.x86_64 is already installed.
Package nmap-ncat-4:7.92-3.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@rishardkogengar:/etc/yum.repos.d#
```

Рис. 4: Установка пакета nmap

Удаление и работа с группами пакетов

```
Freed space: 24 M
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Erasing       : nmap-4:7.92-3.el10.x86_64        1/1
  Running scriptlet: nmap-4:7.92-3.el10.x86_64      1/1

Removed:
  nmap-4:7.92-3.el10.x86_64

Complete!
root@rishardkogengar:/etc/yum.repos.d# dnf remove nmap\*
Dependencies resolved.
=====
Package                Architecture    Version           Repository        Size
=====
Removing:
  nmap-ncat            x86_64         4:7.92-3.el10    @AppStream        470 k

Transaction Summary
=====
Remove 1 Package

Freed space: 470 k
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded
```

Удаление и работа с группами пакетов

```
Last metadata expiration check: 0:02:25 ago on Sun Sep 14 13:21:24 2025.
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Legacy UNIX Compatibility
  Smart Card Support
  Console Internet Tools
  .NET Development
  Graphical Administration Tools
  Network Servers
  RPM Development Tools
  Scientific Support
  Security Tools
  System Tools
root@rishardkogengar:/etc/yum.repos.d# dnf groups info "RPM Development Tools"
Last metadata expiration check: 0:02:56 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Group: RPM Development Tools
Description: Tools used for building RPMs, such as rpmbuild.
Mandatory Packages:
  redhat-rpm-config
  rpm-build
Default Packages:
  rpmdevtools
root@rishardkogengar:/etc/yum.repos.d#
```

```
root@rishardkogengar:/etc/yum.repos.d# dnf history
ID      | Command line                                     | Date and time   | Action(s)      | Altered
-----|-----
 6 | groupremove RPM Development Tools               | 2025-09-14 13:25 | Removed        | 3
 5 | groupinstall RPM Development Tools               | 2025-09-14 13:24 | Install        | 3
 4 | remove nmap*                                    | 2025-09-14 13:23 | Removed        | 1
 3 | remove nmap                                     | 2025-09-14 13:23 | Removed        | 1
 2 | install nmap                                    | 2025-09-14 13:22 | Install        | 1
 1 |                                                  | 2025-09-04 18:16 | Install        | 1373 >E

root@rishardkogengar:/etc/yum.repos.d# dnf history undo 6
Last metadata expiration check: 0:04:11 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing group/module packages:
rpmdevtools            noarch           9.6-9.el10       appstream         87 k
Installing dependencies:
python3-argcomplete     noarch           3.2.2-4.el10     appstream         88 k
Installing Groups:
RPM Development Tools

Transaction Summary
=====
Install 2 Packages
```

Рис. 7: История и откат изменений

```
root@rishardkogengar:/etc/yum.repos.d# dnf list lynx
Last metadata expiration check: 0:06:05 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Available Packages
lynx.x86_64                                2.9.0-6.el10                                appstream
root@rishardkogengar:/etc/yum.repos.d# dnf install lynx --downloadonly
Last metadata expiration check: 0:06:20 ago on Sun 14 Sep 2025 01:21:24 PM MSK.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
lynx                 x86_64            2.9.0-6.el10     appstream         1.6 M

Transaction Summary
=====
Install 1 Package

Total download size: 1.6 M
Installed size: 6.0 M
DNF will only download packages for the transaction.
Is this ok [y/N]: y
Downloading Packages:
lynx-2.9.0-6.el10.x86_64.rpm                6.3 MB/s | 1.6 MB    00:00
-----
Total                                         2.1 MB/s | 1.6 MB    00:00
Complete!

The downloaded packages were saved in cache until the next successful transaction.
You can remove cached packages by executing 'dnf clean packages'.
root@rishardkogengar:/etc/yum.repos.d#
```

Рис. 8: Загрузка пакета lynx

Установка lynx

```
root@rishardkogengar:/etc/yum.repos.d# cd /var/cache/dnf/appstream-8f18e4146890fdcf/packages/
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -Uvh lynx-2.9.0-6.el10.x86_64.rpm
Verifying... ##### [100%]
Preparing... ##### [100%]
Updating / installing...
  1:lynx-2.9.0-6.el10 ##### [100%]
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# which lynx
/usr/bin/lynx
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qf $(which lynx)
lynx-2.9.0-6.el10.x86_64
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qi lynx
Name       : lynx
Version    : 2.9.0
Release    : 6.el10
Architecture: x86_64
Install Date: Sun 14 Sep 2025 01:28:55 PM MSK
Group      : Unspecified
Size       : 6283241
License    : GPL-2.0-only
Signature  : RSA/SHA256, Thu 22 May 2025 11:34:29 PM MSK, Key ID 5b106c736fedfc85
Source RPM : lynx-2.9.0-6.el10.src.rpm
Build Date : Fri 01 Nov 2024 01:03:33 PM MSK
Build Host : pb-913aa2bb-e2a2-4928-a48c-b613e15933c9-b-x86-64
Packager   : Rocky Linux Build System (Peridot) <relog@rockylinux.org>
Vendor     : Rocky Enterprise Software Foundation
URL        : https://lynx.invisible-island.net/
Summary    : A text-based Web browser
Description:
Lynx is a text-based Web browser. Lynx does not display any images,
but it does support frames, tables, and most other HTML tags. One
advantage Lynx has over graphical browsers is speed; Lynx starts and
exits quickly and swiftly displays web pages.
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#
```

Изучение пакета lynx

```
-----  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -ql lynx  
/etc/lynx-site.cfg  
/etc/lynx.cfg  
/etc/lynx.lss  
/usr/bin/lynx  
/usr/lib/.build-id  
/usr/lib/.build-id/69  
/usr/lib/.build-id/69/febb96205311acd4b77a65d52b8e6f7eea76c3  
/usr/share/doc/lynx  
/usr/share/doc/lynx/README  
/usr/share/doc/lynx/docs  
/usr/share/doc/lynx/docs/CHANGES2.3  
/usr/share/doc/lynx/docs/CHANGES2.4  
/usr/share/doc/lynx/docs/CHANGES2.5  
/usr/share/doc/lynx/docs/CHANGES2.6  
/usr/share/doc/lynx/docs/CHANGES2.7  
/usr/share/doc/lynx/docs/CHANGES2.8  
/usr/share/doc/lynx/docs/CRAWL.announce  
/usr/share/doc/lynx/docs/FM.announce  
/usr/share/doc/lynx/docs/IBMPC-charsets.announce  
/usr/share/doc/lynx/docs/README.TRST  
/usr/share/doc/lynx/docs/README.chartrans  
/usr/share/doc/lynx/docs/README.cookies  
/usr/share/doc/lynx/docs/README.defines  
/usr/share/doc/lynx/docs/README.metrics  
/usr/share/doc/lynx/docs/README.options
```

Рис. 10: Список файлов пакета lynx

Изучение пакета lynx

```
rkogengar@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdc/packa...  
/var/cache/dnf/appstream-8f18e4146890fdc/packages  
LYNX(1)                               Lynx - the text web browser                               LYNX(1)  
  
NAME  
    lynx - a general purpose distributed information browser for the World Wide Web  
  
SYNOPSIS  
    lynx [options] [optional paths or URLs]  
  
    lynx [options] [path or URL] -get_data  
    data  
    --  
  
    lynx [options] [path or URL] -post_data  
    data  
    --  
  
    Use "lynx -help" to display a complete list of current options.  
  
DESCRIPTION  
    Lynx is a fully-featured World Wide Web (WWW) client for users running cursor-  
    addressable, character-cell display devices (e.g., vt100 terminals, vt100 emulators  
    running on Windows or any POSIX platform, or any other "curses-oriented" display). It  
    will display hypertext markup language (HTML) documents containing links to files  
    residing on the local system, as well as files residing on remote systems running  
    Gopher, HTTP, FTP, WAIS, and NNTP servers.  
  
    Lynx has been ported to many systems, including all variants of Unix, Windows (since  
    Windows 95), DOS DJGPP and OS/2, as well as VMS and OS/390.  
  
    Lynx can be used to access information on the World Wide Web, or to build information  
    systems intended primarily for local access. For example, Lynx has been used to build  
    several Campus Wide Information Systems (CWIS). In addition, Lynx can be used to build  
    systems isolated within a single LAN.  
Manual page lynx(1) line 1 (press h for help or q to quit)
```

Проверка и удаление lynx

```
Rocky Linux 10                                     Welcome to Rocky Linux

Welcome to Rocky Linux

A community Enterprise Operating System

-----

Rocky Linux is an Enterprise Linux distribution based on the sources available from the
CentOS Project and Red Hat, Inc. used for CentOS Stream and Red Hat Enterprise Linux.
Rocky Linux intends to be binary compatible with the upstream distributions. The source
code can be found at our Rocky Linux Git Service.

Rocky Linux is developed by a diverse team of core developers, with various backgrounds
and skill sets that allow them to build, test, and provide Rocky Linux to the world. The
core developers are supported by an active user community as well, which includes many
testers from the Rocky Linux Testing Team, system administrators, system engineers,
managers, core Linux contributors, and Linux enthusiasts.

Rocky Linux has an active user community with many avenues to talk to, work with, or
support other Rocky Linux users, as well as responsive communication with the Rocky Linux
core development team. There are many ways to join and take part in the community:
Mattermost, IRC/Matrix, Forums, Email Lists, Reddit

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Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```



```
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# man lynx  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qc lynx  
/etc/lynx-site.cfg  
/etc/lynx.cfg  
/etc/lynx.lss  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -q --scripts lynx  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -e lynx  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# ls  
lynx-2.9.0-6.el10.x86_64.rpm  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# █
```

Рис. 13: Конфигурационные файлы lynx

```
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# dnf list dnsmasq
Last metadata expiration check: 0:04:40 ago on Sun 14 Sep 2025 01:29:22 PM MSK.
Installed Packages
dnsmasq.x86_64                                2.90-4.el10                                @AppStream
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# dnf install dnsmasq
Last metadata expiration check: 0:04:48 ago on Sun 14 Sep 2025 01:29:22 PM MSK.
Package dnsmasq-2.90-4.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# which dnsmasq
/usr/sbin/dnsmasq
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qf $(which dnsmasq)
dnsmasq-2.90-4.el10.x86_64
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages#
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdcf/packages# rpm -qi dnsmasq
Name       : dnsmasq
Version    : 2.90
Release    : 4.el10
Architecture: x86_64
Install Date: Thu 04 Sep 2025 06:17:53 PM MSK
Group      : Unspecified
Size       : 794976
License    : GPL-2.0-only OR GPL-3.0-only
Signature  : RSA/SHA256, Thu 22 May 2025 03:17:11 AM MSK, Key ID 5b106c736fedfc85
Source RPM : dnsmasq-2.90-4.el10.src.rpm
Build Date : Thu 31 Oct 2024 02:46:28 PM MSK
Build Host : pb-6df20f45-04b4-477b-bded-5a041ce1b59d-b-x86-64
```

Рис. 14: Установка и определение расположения dnsmasq

```
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdtc/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdtc/packages# rpm -ql dnsmasq  
/etc/dbus-1/system.d/dnsmasq.conf  
/etc/dnsmasq.conf  
/etc/dnsmasq.d  
/usr/lib/.build-id  
/usr/lib/.build-id/c9  
/usr/lib/.build-id/c9/a4bb4576977dbef150b40955510f869d246b34  
/usr/lib/systemd/system/dnsmasq.service  
/usr/lib/sysusers.d/dnsmasq.conf  
/usr/sbin/dnsmasq  
/usr/share/dnsmasq  
/usr/share/dnsmasq/trust-anchors.conf  
/usr/share/doc/dnsmasq  
/usr/share/doc/dnsmasq/CHANGELOG  
/usr/share/doc/dnsmasq/DBus-interface  
/usr/share/doc/dnsmasq/FAQ  
/usr/share/doc/dnsmasq/doc.html  
/usr/share/doc/dnsmasq/setup.html  
/usr/share/licenses/dnsmasq  
/usr/share/licenses/dnsmasq/COPYING  
/usr/share/licenses/dnsmasq/COPYING-v3  
/usr/share/man/man8/dnsmasq.8.gz  
/var/lib/dnsmasq  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fdtc/packages# █
```

Рис. 15: Список файлов пакета dnsmasq

DNSMASQ(8)

System Manager's Manual

DNSMASQ(8)

NAME

dnsmasq - A lightweight DHCP and caching DNS server.

SYNOPSIS

dnsmasq [OPTION]...

DESCRIPTION

dnsmasq is a lightweight DNS, TFTP, PXE, router advertisement and DHCP server. It is intended to provide coupled DNS and DHCP service to a LAN.

Dnsmasq accepts DNS queries and either answers them from a small, local, cache or forwards them to a real, recursive, DNS server. It loads the contents of /etc/hosts so that local hostnames which do not appear in the global DNS can be resolved and also answers DNS queries for DHCP configured hosts. It can also act as the authoritative DNS server for one or more domains, allowing local names to appear in the global DNS. It can be configured to do DNSSEC validation.

The dnsmasq DHCP server supports static address assignments and multiple networks. It automatically sends a sensible default set of DHCP options, and can be configured to send any desired set of DHCP options, including vendor-encapsulated options. It includes a secure, read-only, TFTP server to allow net/PXE boot of DHCP hosts and also supports BOOTP. The PXE support is full featured, and includes a proxy mode which supplies PXE information to clients whilst DHCP address allocation is done by another server.

The dnsmasq DHCPv6 server provides the same set of features as the DHCPv4 server, and in addition, it includes router advertisements and a neat feature which allows naming for clients which use DHCPv4 and stateless autoconfiguration only for IPv6 configuration. There is support for doing address allocation (both DHCPv6 and RA) from subnets which are dynamically delegated via DHCPv6 prefix delegation.

Manual page dnsmasq(8) line 1 (press h for help or q to quit)

Конфигурация и скрипты dnsmasq

```
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -qd dnsmasq  
/usr/share/doc/dnsmasq/CHANGELOG  
/usr/share/doc/dnsmasq/DBus-interface  
/usr/share/doc/dnsmasq/FAQ  
/usr/share/doc/dnsmasq/doc.html  
/usr/share/doc/dnsmasq/setup.html  
/usr/share/man/man8/dnsmasq.8.gz  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# man dns masq  
No manual entry for dns  
No manual entry for masq  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# man dnsmasq  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages#  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -qc dnsmasq  
/etc/dbus-1/system.d/dnsmasq.conf  
/etc/dnsmasq.conf  
root@rishardkogengar:/var/cache/dnf/appstream-8f18e4146890fd9c/packages# rpm -q --scripts dnsmasq  
preinstall scriptlet (using /bin/sh):  
#precreate users so that rpm can install files owned by that user  
  
# generated from dnsmasq-systemd-sysusers.conf  
getent group 'dnsmasq' >/dev/null || groupadd -r 'dnsmasq' || :  
getent passwd 'dnsmasq' >/dev/null || \  
    useradd -r -g 'dnsmasq' -d '/var/lib/dnsmasq' -s '/usr/sbin/nologin' -c 'Dnsmasq DHCP and DNS  
server' 'dnsmasq' || :  
postinstall scriptlet (using /bin/sh):
```

Рис. 17: Установочные скрипты dnsmasq

Итоги работы

В ходе работы были закреплены практические навыки управления пакетами в Linux с помощью **dnf** и **rpm**:

- работа с репозиториями и группами пакетов,
- установка, удаление и изучение содержимого пакетов,
- использование конфигурационных файлов и скриптов,
- контроль безопасности устанавливаемых пакетов.